Web Development

>> AJAX

Asynchronous JavaScript and XML

This lecture is based on lectures prepared by Mohamed Misbhauddin in Advanced Web-based Systems

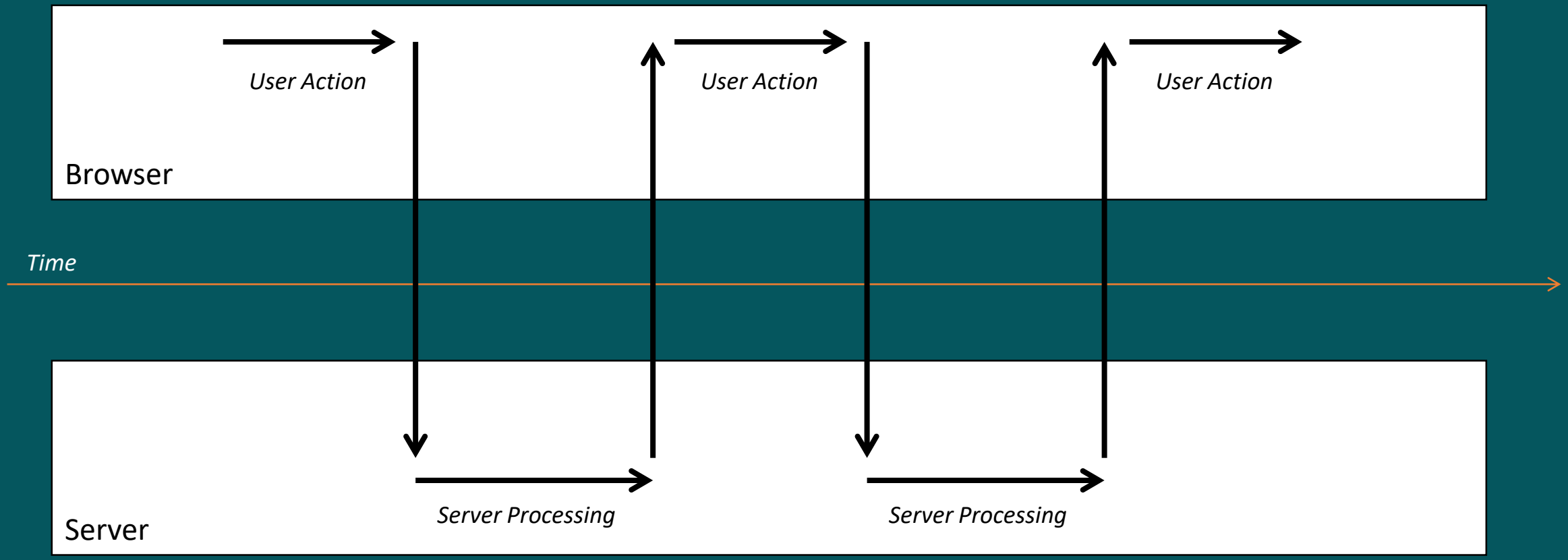
A synchronous

J

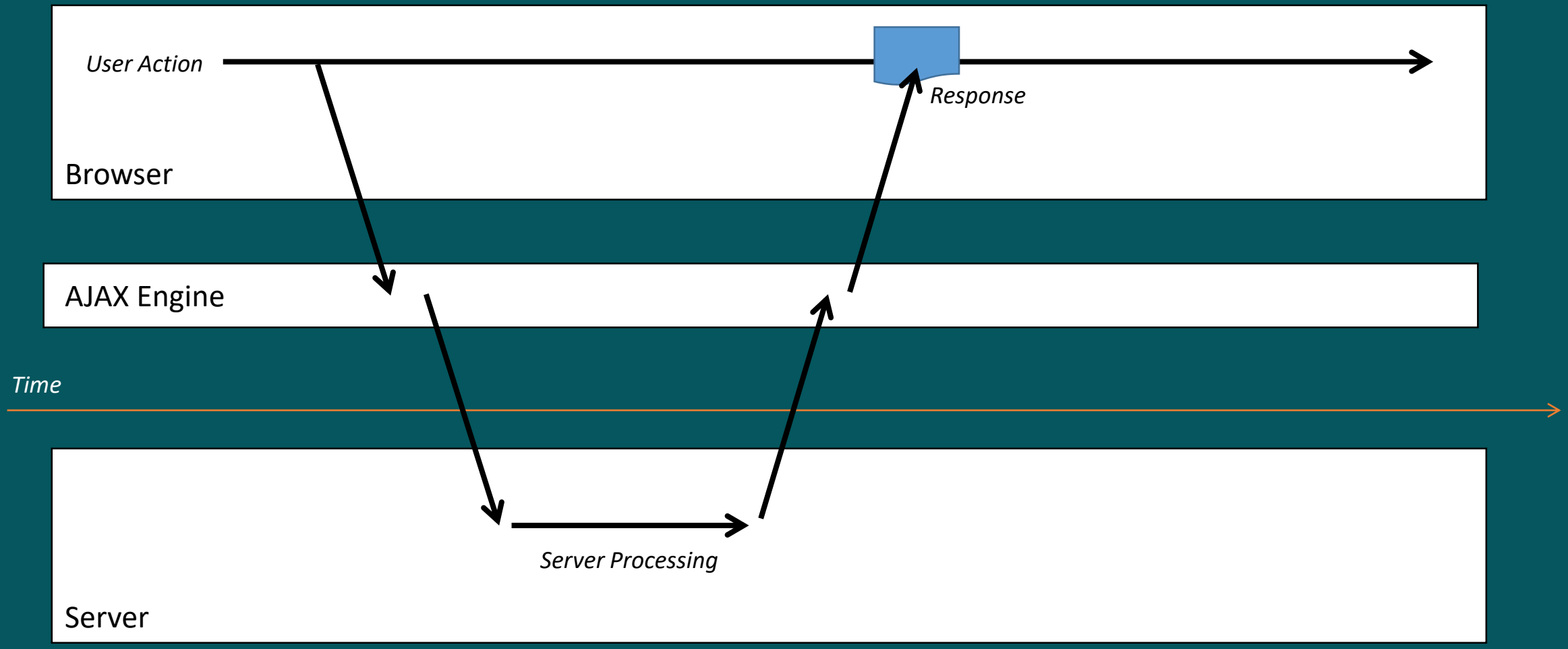
A

X

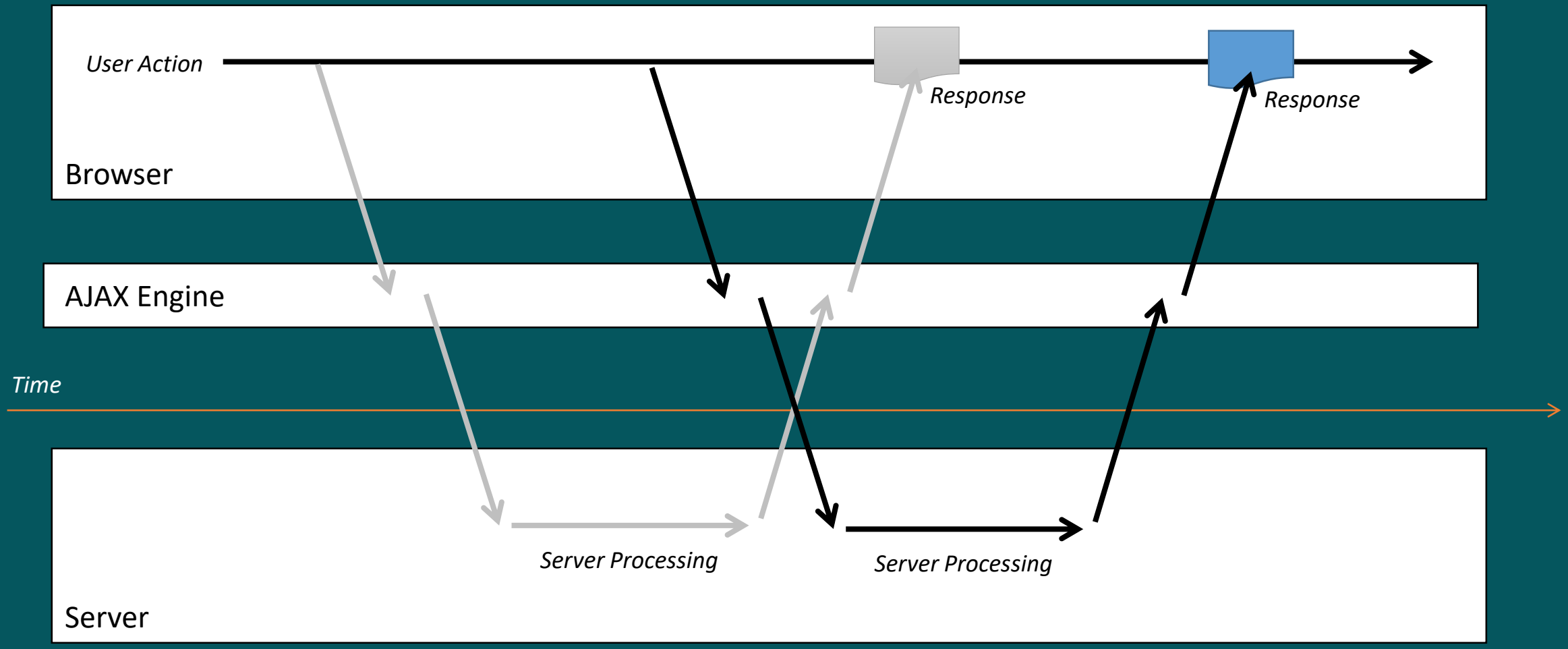
Synchronous Call



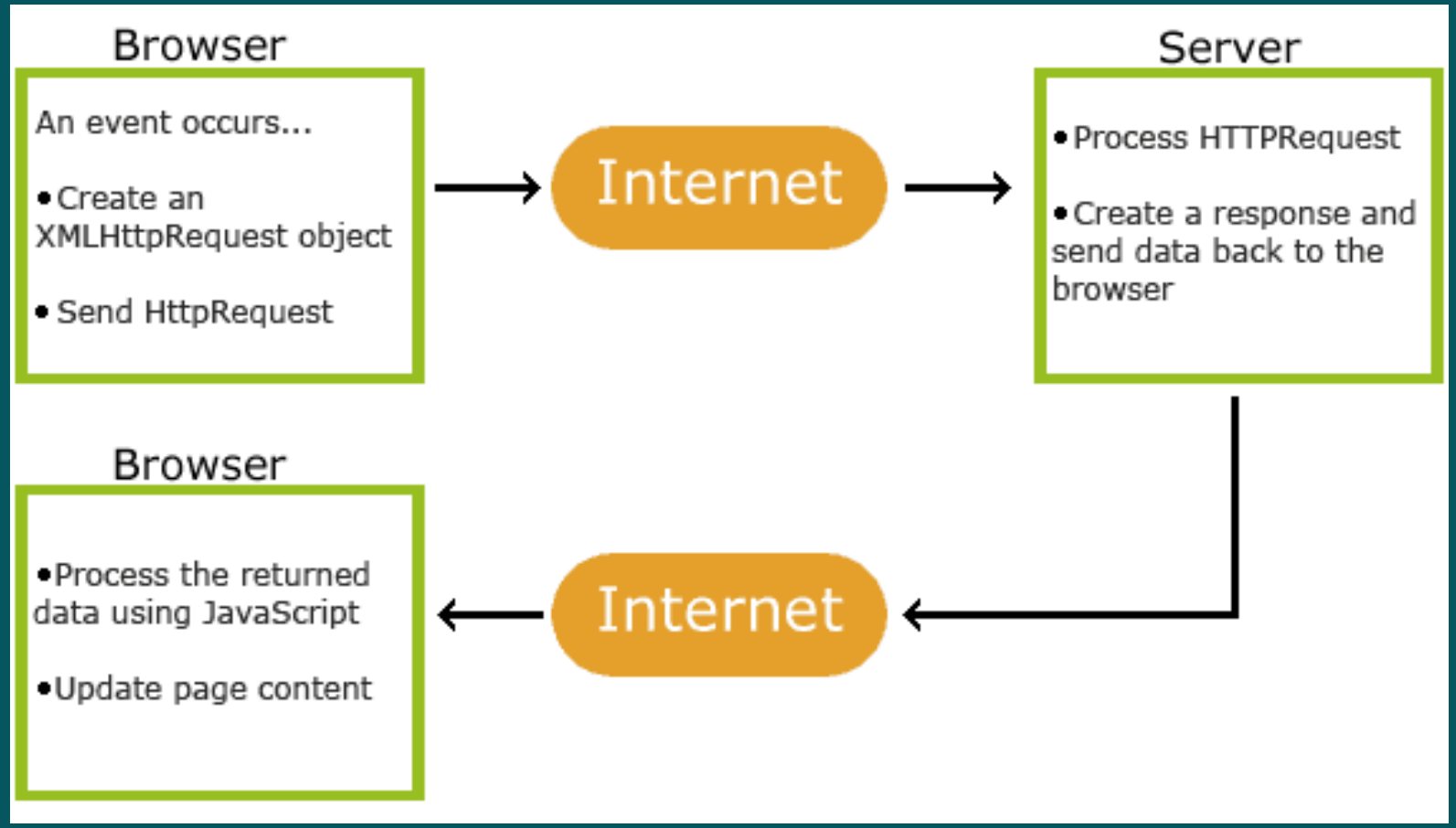
Asynchronous Call



Asynchronous Call



How AJAX Works



https://www.w3schools.com/xml/ajax_intro.asp

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JavaScript Objects

- Use the XMLHttpRequest (XHR) Object
  - Used to exchange data asynchronously with a server
  - It update parts of a web page, without reloading the whole page
  - It is the keystone of AJAX

XHR Object

Attributes

- readyState
- responseText
- status
- statusText

Events

- onreadystatechange
- onload
- onloadstart
- onprogress

Methods

- open
- send
- abort

Asynchronous

Javascript

And

XML

Response of an AJAX Call

- XML
  - Used in earlier days
- JSON (JavaScript Object Notation)
  - Now the most popular data standard used
- HTML
  - Also used popularly
  - Mainly when the data sent from the server is for display as-is

AJAX Step-by-Step

STEP 1: Create XHR Object

```
var xhr = new XMLHttpRequest();
```

- **xhr** is the name of the object {any name can be used}
- **new** is the keyword used to create a **new Object**
- **XMLHttpRequest** is the name of the **object's class**

AJAX Step-by-Step

STEP 2: Create a Callback Function

```
xhr.onreadystatechange = function () {


}
```

- **onreadystatechange** event is fired every time there is a ***change*** in the event
- using anonymous function for call back (remember JavaScript class)

- Create new **folder** and make new **html file** in it
- Put **h1**, **button**, **div** in your new page
- Run it from ***localhost*** – after running your webserver(**xampp**)

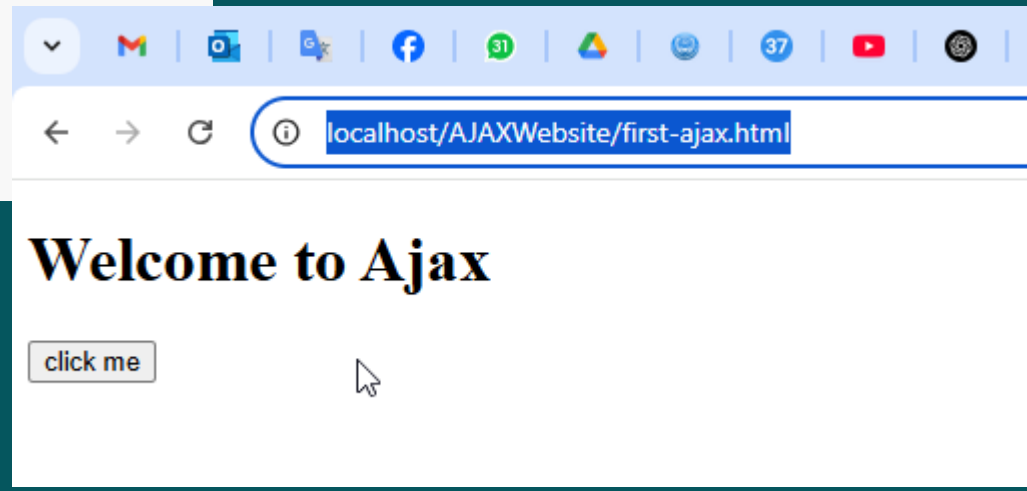
Working Files

first-ajax.html

AJAXWebsite

first-ajax.html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>Document</title>
6  </head>
7  <body>
8      <h1>Welcome to Ajax</h1>
9      <button id="btn">click me</button>
10     <div id="demo"></div>
11 </body>
12 </html>
```



```
3  <head>
4      <meta charset="UTF-8">
5      <title>Document</title>
6  </head>
7  <body>
8      <h1>Welcome to Ajax</h1>
9      <button id="btn" onclick="actionMeth()">click
       me</button>
10     <div id="demo"></div>
11
12     <script>
13     var xhr = new XMLHttpRequest();
14         console.log(xhr);
15         function actionMeth(){
16
17
```

Elements Console Sources Network Performance

top Filter Default levels No Issues

first-ajax.html:14

```
XMLHttpRequest
  onabort: null
  onerror: null
  onload: null
  onloadend: null
  onloadstart: null
  onprogress: null
  onreadystatechange: null
  ontimeout: null
  readyState: 0
  response: ""
  responseText: ""
  responseType: ""
  responseURL: ""
  responseXML: null
  status: 0
  statusText: ""
  timeout: 0
  upload: XMLHttpRequestUpload {onloadstart: null, onprogress: null, ona
  withCredentials: false
  [[Prototype]]: XMLHttpRequest
```

Using xhr to open connection with server

```
}
xhr.open()'GET','getInfo.txt',true);
```

(String method, String url, [bool async], [String user], [String password])

```
<script>
var xhr = new XMLHttpRequest();
    console.log(xhr.readyState);
```

0

```
var xhr = new XMLHttpRequest();
   xhr.open('GET','getInfo.txt',true);
    console.log(xhr.readyState);
```

1

```
<h1>Welcome to Ajax</h1>
<button id="btn" onclick="actionMeth()">click me</button>
<div id="demo"></div>

<script>
var xhr = new XMLHttpRequest();

    function actionMeth(){
        console.log(xhr.readyState);

    }
    xhr.open('GET','getInfo.txt',true);
    xhr.send();

</script>
```

4
Only after press click me

readyState

This variable holds the status of the XMLHttpRequest. Changes from 0 to 4.

0 – Request not initialized
1 – Server connection established
2 – Request Received
3 – Processing Request
4 – Request Finished and response is ready

```
 function actionMeth(){
  //   console.log(xhr.readyState);
xhr.onreadystatechange= function()
    {
   if(xhr.readyState==4 && xhr.status ==200)
     document.getElementById("demo").innerHTML=this.responseText;
}
    |
 xhr.open('GET','getInfo.txt',true);
 xhr.send();
}
```

this = xhr

status

This is the same as the status of the HTTP response codes.

200 – OK
404 – Page not found
403 – "Forbidden"
.....

```
xhr.onreadystatechange= function()
    {
   if(xhr.readyState==4 && xhr.status ==200)
     console.log("found");
 elseif(xhr.status ==404)
    console.log("not found");
}

 }
 xhr.open('GET','getInfo21.txt',true);
 xhr.send();
```

```
GET http://localhost/AJAXWebsite/getInfo21.txt 404 first-ajax.html:27
(Not Found)
    (anonymous) @ first-ajax.html:27
```

AJAX Step-by-Step

STEP 3: Do something when the request completed and response is ready

```
xhr.onreadystatechange = function () {

    if( xhr.readyState == 4  && xhr.status == 200)
    {

    }
}
```

AJAX Step-by-Step

STEP 4: Send the Request to the Server

```
xhr.open("GET",“process.php",true);
xhr.send();
```

- The open method specifies the type of request, the URL, and if the request should be handled
asynchronously or not
- SYNTAX: open(method, url, async)
method: the type of request: GET or POST
url: the location of the file on the server
async: true (asynchronous) or false (synchronous)
- The send method sends the request off to the server
The final step in the request handler is to send the request to the server.
- Has parameters which are used only in POST request

Complete Code

```
var xhr = new XMLHttpRequest();

xhr.onreadystatechange = function () {

    if( xhr.readyState == 4  && xhr.status == 200)
    {
            console.log(xhr.responseText);
    }
}

xhr.open("GET",“process.php",true);
xhr.send();
```

Using synchronous method

```
var xhr = new XMLHttpRequest();
    console.log(xhr.readyState);
     function actionMeth(){
      //   console.log(xhr.readyState);
/*     xhr.onreadystatechange= function()
        {
       if(xhr.readyState==4 && xhr.status ==200)
         document.getElementById("demo").innerHTML=this.responseText;
    }
*/
     xhr.open('GET','getInfo.txt',false);
     xhr.send();
     document.getElementById("demo").innerHTML=xhr.responseText;
    }
```

Most commonly used method(onload)

```
var xhr = new XMLHttpRequest();
   console.log(xhr.readyState);
    function actionMeth(){

  xhr.onload= function(){
      if(this.status ==200)
         document.getElementById("demo").innerHTML=this.responseText;
         //recieve response

  }
  xhr.open('GET','getInfo.txt',true); // connect to server
    xhr.send();// send request
```

AJAX with JQuery

- AJAX with JQuery becomes much simpler
- No need for so many steps as with using XHR Object
- Only one function called $.ajax()
- It includes everything such as the method, url, data and the callback function

$.ajax() function in JQuery

```
$.ajax({
        url: 'process.php',
        type: 'POST',
        crossDomain: true,
        data: { ‘key': value, ……. },
      }).done(function(data) {
          if(data==="success"){

              }else{

              }
      }).fail( function() {
              });
```

Conclusion

- **Data from HTML can be sent to the Server using two basic method GET and POST**
- **Data can be sent using two different ways: synchronous and asynchronous**
- **AJAX is used for asynchronous data transfer**